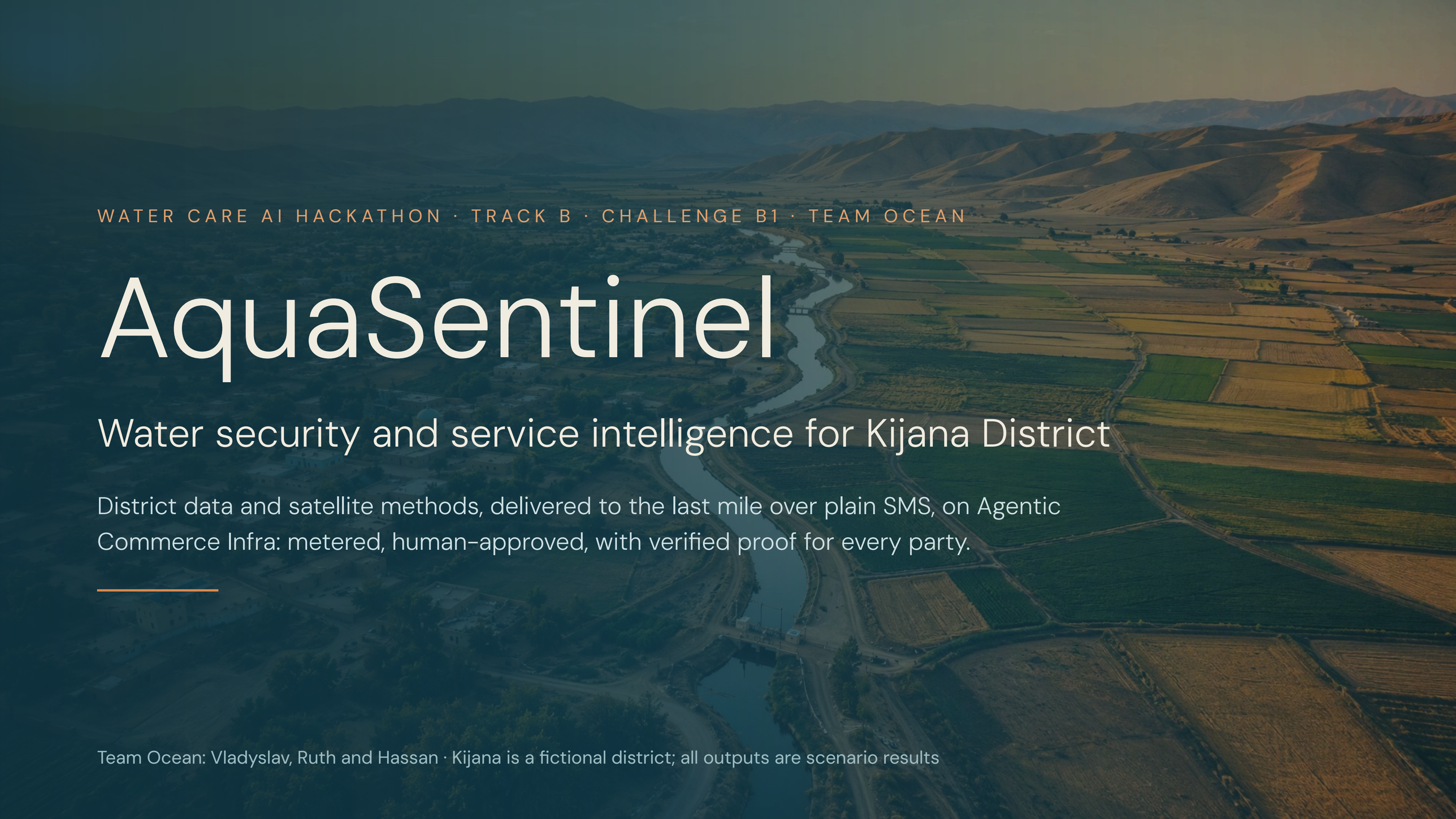


WATER CARE AI HACKATHON · TRACK B · CHALLENGE B1 · TEAM OCEAN

AquaSentinel

Water security and service intelligence for Kijana District

District data and satellite methods, delivered to the last mile over plain SMS, on Agentic Commerce Infra: metered, human-approved, with verified proof for every party.

Team Ocean: Vladyslav, Ruth and Hassan · Kijana is a fictional district; all outputs are scenario results

THE PROBLEM, FROM THE DISTRICT'S OWN DATA

The pumps exist. Nobody knows which ones work, and repairs wait for money.

70%

of 1,078 water points work by the newest evidence. The register says 79%.

593

days: the age of the register's median status record.

84 vs 0

days to repair: district-funded against the pooled service fund.

Kijana District data pack (synthetic scenario) · details in the appendix

WHAT WE BUILT

A list the officer can defend, and money anyone can trace

Score

Every water point out of 100, five parts of 20. Not working always goes first.

Approve

The officer approves, edits or rejects. Nothing goes out without him.

Text

Committees get plain SMS and reply OK, BROKEN, DRY or DIRTY.

Prove

Sponsors fund a water point or the district; every spend has a receipt on Qatom.

Running now on all 1,078 water points. Let us show you.

LIVE DEMO

starwaterheaven.github.io/AquaSentinel

1. Olagoni Shallow Well: 26 out of 100, and why
2. The officer approves; the alert reaches the committee’s phone
3. The committee replies; the score changes
4. A $1 sponsorship, paid live on Qatom this morning, and where it went

Qatom transactions 91ca3e13 ($1 sponsorship) and 9d919d86 ($0.02 approved alert), 17 Oct 2026

DOES IT WORK?

We froze the records at the end of 2023, scored blind, then checked against the 2024 inspections

47%

of the points we flagged were found broken, more than twice the 21% base rate.

72%

of the broken pumps the inspectors found were already on our list.

624

water points inspected in 2024; none of those records fed the score.

scoring/backtest.py on the Kijana data pack (synthetic) · against the Aug 2025 survey of 124 points: 47% precision, 49% recall, 28% base rate

IN REAL LIFE

People decide. Silence is never read as “working”.

Who owns it

The District Water Officer. He approves every alert and repair; committees keep their tariffs and decisions.

Who it could miss

People like Bibi Amina, with no phone. No report or a stale record puts her pump on the inspection round.

What it costs

About $3.50–7.30 per water point per year. The whole district fits in its existing monitoring budget line.

Year three: the district pays for the platform; sponsors fund repairs through escrow, with a receipt for every dollar

BEFORE ANYONE RELIES ON IT

- Check the score against the next inspection round
- Test the SMS, in Swahili, with real committees
- Pilot pump plates and sponsorship in one sub-county

The data exists. The phones exist. We built the list the officer can defend and the money trail anyone can check.

Team Ocean · github.com/StarwaterHeaven/AquaSentinel · starwaterheaven.github.io/AquaSentinel

APPENDIX

Detail for questions

Evidence · which number we trust · system map · data · how the score works · product · people · governance · cost · what is live · team · build and testing

THE STAKES · KIJANA DISTRICT

420,000 people, about one in three water points down

~1,042

water points on the district register

~65%

actually functional, against the 78.9% the district reports

970

volunteer water committees keeping them running

13 + 3

mechanics and technicians for the whole district, 88% rural

Globally, 2.1 billion people still lack safely managed water. In Kijana the gap is not detection: it is keeping the pumps that already exist working.

Sources: Kijana District B1 case pack and data pack (synthetic scenario) · WHO/UNICEF JMP 2025

THE PROBLEM

Three gaps keep Kijana’s water points failing

01

No early warning

The register’s median status record is **593 days old**; 43% are over two years. The district’s own 78.9% is stale on its face.

02

Response lag is a money problem

Median repair: **84 days** district-funded, 7 days committee-funded, 0 days from the service-agreement pool. Repairs cluster in the Oct, Jan and Apr budget-release months.

03

The last-mile gap

Reporting volume halved when the pilot’s staffing funds ended in mid-2023. Two-thirds of Mlimani has no signal; some users have no phone at all.

Source: Kijana water-point register, job cards and community messages (B1 data pack, synthetic)

THE EVIDENCE · WHICH NUMBER WE TRUST

Three answers to “how many water points work?”

79%

The register. 851 of 1,078 rows say Functional, but the median record is 593 days old.

70%

Newest evidence. 755 working after we take the latest of register, inspections, survey, job cards and committee SMS.

~65%

The scenario’s working figure. Consistent with ours once silent points are counted.

We trust 70% as a ceiling. 381 of the points we count as working have no fresh evidence at all. Until they are checked, the true figure sits between 65% and 70%.

Source: AquaSentinel scoring on the Kijana data pack (synthetic), status as of 30 Sep 2026

OUR ANSWER

Turn the district’s own records and climate signals into a weekly water-security score for every village, and push it to any phone, even without internet.

One pipeline. One score. Two outputs: a dashboard for officers, SMS for committees. And Agentic Commerce Infra underneath: every agent action metered, receipted and human-approved, and every dollar traceable.

SYSTEM MAP

From district record to delivered warning

1 · Observe

Kijana data pack: water-point register, job cards, community messages, 5 km rainfall grid, groundwater, flood extents, settlements, catchments.

2 · Enrich

Pipeline cleans and joins the files; synthetic records fill gaps, always flagged.

3 · Score

0–100 water-security score per village and water point, updated weekly.

4 · Reach

Plain SMS in Swahili to committees; priority list to officers.

5 · AGENTIC COMMERCE INFRA

Every stage is a metered tool call. Broadcasts and dispatches wait at an officer's approval link. Every action and every dollar leaves verified proof.

Dashboard for district water officers and NGOs: map, trends, the weekly priority list.

SMS report for water committees: one actionable message per week.

DATA LAYER · DATA ENGINEERING

The district’s data drives it; public feeds check the method

Kijana district data (primary)

Data pack file
Water-point register and status history
Repair job cards
Community messages
5 km rainfall grid
Groundwater monitoring
Flood extents
Settlements and catchment links

Public feeds (methods only)

Source	Validates
CHIRPS / ERA5	Rainfall anomaly method
Sentinel-2	Surface water, vegetation
SMAP / GRACE	Soil and groundwater trend
FAO AQUASTAT	Withdrawal baselines
OSM + HDX	Water-point schema

Rules of use

Public sources do **not** align with Kijana’s invented geography. They validate methods and benchmarks; they never feed the fictional district.

Where data is missing, synthetic records fill the gap, tagged **data_flag = SYNTHETIC** at ingestion and never mixed silently with observed data.

HOW THE SCORE WORKS

One score out of 100, five parts of 20, no black box

Working now

Newest report from any source. Old reports count for less.

Reliability

Breakdowns since 2021 and how long repairs take.

Committee

Tariff, savings, recent meeting, service agreement.

Water quality

Latest E. coli, fluoride, nitrate; dirty-water reports.

Water supply

Rain against normal; shallow wells weigh more; dry reports.

Rule: not working goes first

Any point that is not working is flagged and tops the officer's list, whatever its score. 323 today.

Confidence beside every score

No fresh evidence means "silent", and silent points go on the inspection list. 484 today.

Bands: 80+ Good · 60–79 Watch · 40–59 At risk · under 40 Critical · scenario results; a score never shows water is safe to drink

PRODUCT · FRONTEND / UX

Two interfaces, one score, one approval

Officer dashboard

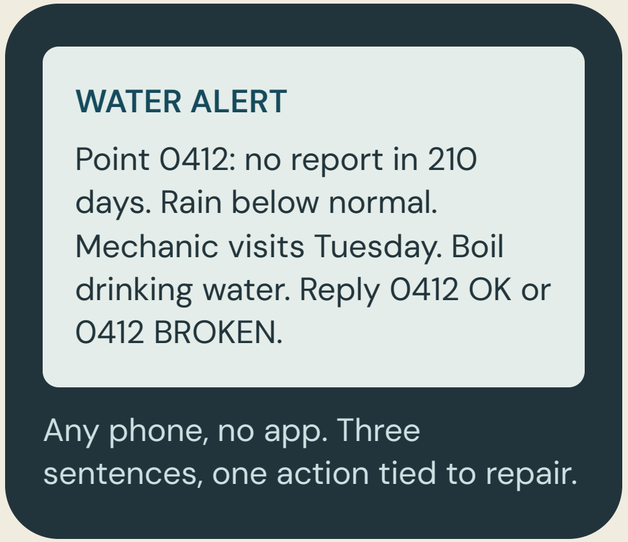
Village map by score, trends, and the weekly priority list.

The list waits in an approval state. The officer accepts, edits or overrides before anything broadcasts, and each decision is recorded.

The silence model

“Communities we haven’t heard from”: no messages in 12 months, a stale register status and a remote settlement put a water point on the inspection list.

Absence of a report is not evidence a pump works.



Any phone, no app. Three sentences, one action tied to repair.

SMS is a mock, shown in English for the demo; in the field it is sent in Swahili first · water point ID illustrative

HOW IT WORKS IN REAL LIFE

Who uses it, who decides, and who it could miss

Joseph, District Water Officer

Owns the list. Approves, edits or rejects every alert and dispatch; nothing goes out without him. No new data entry.

Halima, area mechanic

Gets an approved job card by SMS, with the failed part where known. No smartphone needed; payment from escrow is v1.1.

Mama Rehema, committee treasurer

Plain SMS on a basic phone. Replies OK, BROKEN, DRY or DIRTY and gets an answer, plus her water point’s funds.

Bibi Amina, no phone, Mlimani

The silence list: no report, stale status or no signal puts her pump on the inspection round. Absence of a report is never read as “working”.

Personas from the B1 case pack · 127 settlements have no signal at all; the plate on each pump lets anyone report with its 4-digit number

GOVERNANCE · DOMAIN / PM

Trust is the real infrastructure

1 · Open data in, open methods out

Weights and thresholds are published, so results can be audited and reproduced.

2 · Minimal personal data

Village-level scores and opt-in phone numbers only. No names, no land records.

3 · Humans stay in the loop, by architecture

The broadcast physically waits at the officer's approval link. The model advises; accountable people decide.

4 · Synthetic data is flagged

Labelled at ingestion. Model cards state where training data ends and reality begins.

5 · Every action has a receipt

Metered calls, a send audit, and live funding proof: sponsors see where their money went, the officer sees district funding, each committee sees its village's funds.

COST AND WHO PAYS

About $3.50 to $7.30 per water point per year

Per water point per year	USD
52 weekly alerts at $0.02	$1.04
About 24 reports in and replies out	$0.96
Scoring calls at $0.001	$0.08
SMS shortcode, shared by 1,042 points	$1.15–3.45
Dashboard hosting, shared	$0.23–1.73
Total	**$3.46–7.26**

Year three

Platform: the whole district costs $3,600–7,600 a year, inside the district's own monitoring line ($7,300 budgeted, $5,200 released in 2025/26).

Repairs: sponsors, from NGOs to the public, fund the district or one water point's escrow in real time; committee tariffs top it up. Every spend has a receipt.

Unit costs from the case pack (SMS $0.01–0.03, shortcode $100–300/month, hosting $20–150/month); scoring price is our metered rate

WHAT IS READY NOW

What is live today, and what is not yet

We can deploy this today

- MCP tool interface
- Metered, priced per-call execution
- Wallet and approval links
- Transaction records
- Escrow accounts

v1.1, not yet live

- Community ground-truth data listings
- Automated distribution and scheduling
- Escrow-settled mechanic payments in a live pilot

Scenario rules

- Kijana is fictional; outputs are scenario results
- No model output establishes water is safe to drink
- High-risk alerts carry boil or sachet advice until a lab test says otherwise

TEAM OCEAN

Three people, one open repo

Vladyslav

Data and validation: the backtest and the better message reader. Presenting today.

Ruth

Field and community: the SMS flow, the pump plate, the silence list and who it protects.

Hassan

Product and the Agentic Commerce Infra: scope, funding and approval flows on Qatom.

Built with an AI build agent. Code, contract and data rules are public: github.com/StarwaterHeaven/AquaSentinel

THE BUILD, AND WHAT NEEDS TESTING

Running today on the Kijana data

In the demo

1. 1,078 water points scored; 311 flagged Not working
2. Officer approves Olagoni's alert; the SMS lands on the phone
3. The committee replies; the score changes live
4. The flood messages import and rescore the district
5. Real Qatom receipts from today: $1 sponsorship and a $0.02 approved alert

Before anyone relies on it

- Backtest the score against the next inspection round
- Test SMS wording in Swahili with committees
- Pilot the pump plates and phone registration in one sub-county
- Pilot sponsorship escrow with one district water office

The data exists. The phones exist. What was missing is a list the officer can defend and money anyone can trace. We built both.

Qatom transactions 91ca3e13 (item 212, $1) and 9d919d86 (item 213, $0.02), 17 Oct 2026 · try it: starwaterheaven.github.io/AquaSentinel